

Python Programming & Business Analytics

Course Syllabus

Term: Fall 2026

Class Time: Thursdays, 3 Hours

Format: IA-Enhanced Learning

Location: Weber 103

Course Description & Philosophy

This semester, we will work together to build and exercise our Python skills and comfortability. This is an interactive, IA-enhanced course designed to scale your technical capacity from foundational code mechanics up to real-world data analytics implementations.

We will actively integrate artificial intelligence as a supportive peer, co-coder, and instructional assistant to help accelerate debugging, explore alternate data structures, and personalize learning paths after all foundational skills are met.

Weekly Schedule & Milestone Timeline

Phase 1: Core Python Foundations

- **Week 1: Course Orientation & Tech Set-Up**
Focus: Introduction to the IA-enhanced framework. Setting up IDE of instructor choice (Jupyter Notebook, VS Code, Spyder, PyCharm), and managing environment packages.
- **Week 2: Murach's Python Programming – Section 1: Part A**
Topic: Essential Concepts and Skills.
Facilitator: Shyla C. Solis
- **Week 3: Murach's Python Programming – Section 1: Part B**
Topic: Deep dive into operations, conditional logic, and control loops.
Facilitator: Shyla C. Solis

Phase 2: Advancing Python Skills

- **Week 4: Murach's Python Programming – Section 2: Part A**
Topic: Other Concepts and Skills (Functions, lists, tuples, and dealing with exceptions).
Facilitator: Wendy Chen
- **Week 5: Murach's Python Programming – Section 2: Part B**
Topic: Dictionaries, strings, and working with dates/times.
Facilitator: Wendy Chen

Phase 3: Applications, Interfaces, & Data Entry

- **Week 8: Murach's Python Programming – Section 4: Part A**
Topic: Database Programming (SQLite/MySQL connectivity and handling data tables).
Facilitator: Xavier Ward
- **Week 9: Murach's Python Programming – Section 4: Part B**
Topic: GUI Programming.
Facilitator: Xavier Ward

Phase 4: Transitioning to Data Science

- **Week 10: Murach's Python for Data Science – Section 1: Part A**
Topic: Get Off to a Fast Start (Introduction to Pandas, NumPy, and essential data frames).
Facilitator: [Sign up available]
- **Week 11: Murach's Python for Data Science – Section 1: Part B**
Topic: Data cleaning, filtering rows, and basic plot structures.
Facilitator: [Sign up available]

Phase 5: Analytical Case Studies

- **Week 12: Case Study – The Polling Case Study: Part A**
Topic: Data ingestion and aggregations of statistical polling datasets.
Facilitator: [Sign up available]
- **Week 13: Case Study – The Polling Case Study: Part B**
Topic: Parsing, analyzing trends, and compiling reports.
Facilitator: [Sign up available]
- **Week 14: Thanksgiving Break (No Class)**
Independent Study: Use this window to refine final project scripts.
- **Week 15: AI inside IDE**
Topic: Learn to speed up production using models in our integrated development environments.
Facilitator: Shyla Solis